STAIR: A Five-Level Framework for Evaluating LMS Readiness for Agentic AI Teaching Agents

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Abstract. Autonomous AI teaching agents promise personalised instruction at scale, but the learning management systems that would host them were designed for human workflows, not agentic ones. This paper shows the mismatch is not a matter of capability but of architecture. Applying a new five-level readiness framework, STAIR (Scaffold for Teaching Agent Integration Readiness), to Moodle 5.2dev+ reveals a 28-percentage-point cliff between student-state infrastructure (53%) and agent-reasoning infrastructure (25%). The cause is code-level isolation between the platform's AI and analytics subsystems. STAIR derives its levels from converging requirements in Zimmerman's [1] self-regulated learning cycle, Talebirad and Nadiri's [18] multi-agent collaboration architecture, and Chu et al.'s [19] educational LLM agent taxonomy. The framework operationalises these into 28 sub-dimensions scored against an evidence-linked rubric through systematic source-code analysis. Moodle scores 55% at event sensing (L1) and 53% at student state (L2) but falls to 25% at agent reasoning (L3), 5% at orchestration (L4), and 22% at intervention delivery (L5). The platform provides no native support for autonomous agent coordination or pedagogically timed response. What prevents AI teaching agents from achieving learning outcomes is not the intelligence of the models but the architecture of the platforms that host them. Until learning management systems bridge language-model reasoning and learner data, the field's growing investment in educational AI remains structurally stranded, with powerful models held back by the platforms rather than by their own limits.

Keywords: *STAIR framework, learning management systems, AI teaching agents, self-regulated learning, agent architecture, Moodle, platform readiness*

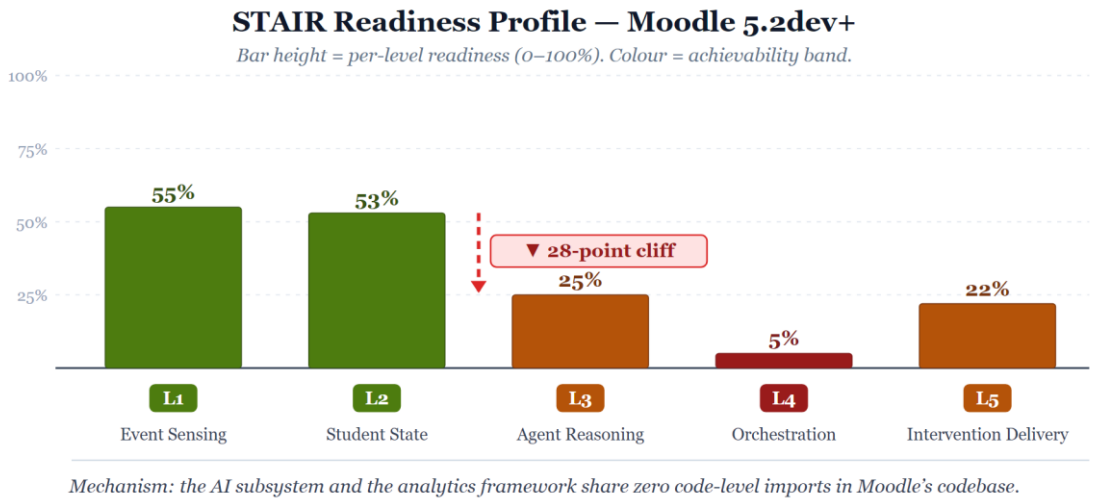

FIGURE 1. STAIR readiness profile for Moodle 5.2dev+. Bar heights report per-level readiness (0–100%); colour encodes the achievability band (green ≥ 40%, amber 20–39%, red < 10%). The 28-percentage-point discontinuity between L2 (53%) and L3 (25%) is the paper's central finding; its mechanism (no code-level connectivity between the AI subsystem and the analytics framework) is unpacked in Section 5.

1. Introduction

Autonomous AI teaching agents are now technically constructible. Large language models can reason about student performance, decompose learning goals into manageable steps, and generate personalised guidance at scale. Yet the learning management systems (LMSs) where students submit assignments, attempt quizzes, and engage in discussions were built to deliver content and record activity, not to host autonomous software that perceives, reasons, and acts on behalf of stakeholders to maximise learning outcomes. Hosting such an agent requires a platform that can sense a student's ongoing activity, maintain their accumulated learning trajectory, reason over that state within pedagogical constraints, coordinate with other agents acting on the same learner, and deliver interventions back into the learning flow. An LMS missing any of these five capabilities cannot host an autonomous teaching agent. The capability gap is not in the AI but in the platform, regardless of how capable the model itself may be.

This mismatch is architectural, not configurational. Configuration changes such as enabling a plugin, activating an API key, or adjusting a permission operate within a platform's existing design. Architectural gaps are structural: they concern whether the event infrastructure can stream learner actions to external

consumers, whether student-state data is exposed through an interface agents can query programmatically, and whether the platform provides any mechanism for coordinating multiple agents acting on the same learner simultaneously. These conditions cannot be patched; they must be designed in.

The reason this gap has not been systematically examined is that the problem is recent. Until the past two to three years, AI teaching systems operated alongside LMS platforms as standalone applications, not as autonomous agents embedded within them. Prior reviews of AI in education have catalogued application areas and deployment challenges [7], [9], [10] but have not examined the platform-side architectural conditions these applications require. The result is a field that understands AI's educational potential but lacks any instrument for determining whether its deployment infrastructure is ready to realise it.

This paper addresses that gap through two research questions:

> ***RQ1: What architectural conditions must a learning management system satisfy to support autonomous AI teaching agents, and how can those conditions be operationalised into a replicable assessment instrument?***

> ***RQ2: To what extent does Moodle 5.2dev+ satisfy those conditions at each architectural level?***

To address these questions, this paper develops STAIR (Scaffold for Teaching Agent Integration Readiness), a five-level diagnostic framework grounded in three converging theoretical pillars: Zimmerman's [1] self-regulated learning cycle, which establishes what agents must observe and respond to; Talebirad and Nadiri's [18] multi-agent collaboration framework, which establishes what agents need from their host environment; and Chu et al.'s [19] educational LLM agent taxonomy, which establishes what educational agents specifically require. The framework is applied to Moodle 5.2dev+ (Build 20260320) through systematic source-code analysis, scoring each architectural level against an evidence-linked rubric with full traceability to the implementation.

This paper makes three contributions:

1. **The STAIR framework:** a five-level diagnostic instrument for evaluating whether an LMS can structurally support autonomous AI teaching agents, grounded in SRL theory and agentic AI research, and applicable to any LMS platform.
2. **An empirical readiness profile of Moodle 5.2dev+:** the first source-code-based application of STAIR to a major open-source LMS, revealing substantial readiness at the data levels (L1: 55%, L2: 53%) and a structural collapse at the reasoning and coordination levels (L3: 25%, L4: 5%, L5: 22%).
3. **Identification of the AI-analytics isolation as the critical barrier:** the L2-to-L3 discontinuity is produced by complete code-level separation between Moodle's AI subsystem and its analytics framework, a structural condition that prevents any agent from connecting language-model capability to learner context.

2. Related Work

Research on autonomous teaching agents spans two literatures that have rarely intersected. Intelligent tutoring systems and self-regulated learning theory describe what such agents must do for learners, focusing on state diagnosis, feedback adaptation, and metacognitive scaffolding. LLM-agent architecture and LMS data infrastructure research turns the question around. It specifies what agents need from a host platform, including event streams, a unified learner state, orchestration, and a delivery channel back to the learning flow. Neither literature asks whether any current LMS can structurally carry an autonomous teaching agent. No instrument exists to test it.

2.1 Educational AI and the SRL Foundation

Research on AI in education has established a rich taxonomy of application areas but has analysed them in isolation from the platform infrastructure on which they depend. Zawacki-Richter et al. [7], reviewing 146 studies, identified four established application areas: profiling and prediction, assessment, adaptive personalisation, and intelligent tutoring. Zerkouk et al. [8], reviewing 127 studies of AI-based intelligent tutoring systems between 2010 and 2025, catalogued persistent challenges in adaptive learning, student modelling, and real-world deployment without examining the platform conditions under which those systems operate. Roll and Wylie [9] argued for a shift toward a "revolutionary process" of embedding AI within students' everyday learning without addressing what platform architecture makes that embedding structurally possible. Kasneci et al. [10], mapping the opportunities and challenges of large language models in education, identified "integration with existing educational ecosystems" as a key challenge but treated the LMS as a passive host rather than an active architectural constraint.

Self-regulated learning theory and intelligent tutoring research establish what an agent must observe to be pedagogically effective. Zimmerman [1] defined SRL as three cyclical phases: forethought (goal-setting and planning), performance (execution and self-monitoring), and self-reflection (evaluation and calibration). Pintrich [2] proposed a parallel four-phase framework organising self-regulation across cognition, motivation, behaviour, and context, and Schunk and Greene [3] synthesised subsequent developments across both traditions, confirming that the multi-dimensional monitoring requirement these frameworks

define remains the theoretical foundation for any system claiming to support SRL. VanLehn [4], in a comprehensive review of tutoring effectiveness, found that intelligent tutoring systems achieved effect sizes (d = 0.76) approaching human tutoring (d = 0.79), substantially higher than answer-based computer-assisted instruction (d = 0.30). Kulik and Fletcher [5], in a meta-analysis of 50 controlled evaluations, confirmed these findings with a median effect size of 0.66 standard deviations. Azevedo et al. [6] demonstrated that real-time scaffolding of learner state produces meaningful metacognitive gains; critically, the MetaTutor system operated as a standalone hypermedia application, not embedded within an LMS, and assumed direct control over the learning environment rather than mediated access through a platform API. The educational AI literature establishes what autonomous agents can achieve. Self-regulated learning and tutoring research establishes what learner states those agents must track to achieve it. Neither specifies what platforms must structurally provide.

2.2 Agent Architecture and LMS Infrastructure

The LLM agent literature makes explicit what autonomous agents require from their operating environment. Xi et al. [11] and Wang et al. [12] decomposed agent architecture into perception, memory, planning, and action modules, each placing specific demands on the host environment: access to learner identity and history, persistent and queryable state storage, real-time contextual input, and an execution interface through which the agent can intervene. The reasoning foundations have been clarified by recent work: Wei et al. [13] demonstrated that chain-of-thought prompting enables multi-step reasoning in large language models, and Huang et al. [14] categorised planning approaches into task decomposition, plan selection, external module integration, and reflection, each imposing distinct requirements on the host environment's ability to provide structured feedback after each reasoning step. Park et al. [15] demonstrated that long-horizon coherent agent behaviour depends on a memory stream, a reflection mechanism, and a planning module, all of which require the host to supply continuous structured event data. Yao et al. [16] showed that the ReAct paradigm, which interleaves reasoning traces with task-specific actions, requires an environment capable of returning an observation after each agent action, a perception-action loop that learning management systems were not designed to provide natively. For multi-agent settings, Dillenbourg [17] identified situation, interaction, mechanism, and effect as the four dimensions of collaborative learning, and Talebirad and Nadiri [18] demonstrated that LLM-based agents cannot collaborate when operating in isolation, making the host platform's coordination infrastructure a prerequisite. Chu et al. [19] synthesised these requirements in an educational context, identifying six core capabilities for educational LLM agents (memory, tool use, planning, personalisation, explainability, and multi-agent communication) and naming "integration with existing educational ecosystems" as the central unsolved challenge.

Research on LMS data infrastructure recognises the fragmentation problem but addresses it through pipeline design rather than readiness diagnosis. Siemens and Baker [20] established the foundational principle that data about learning should serve learning, setting the agenda for the learning analytics field. Thajchayapong et al. [21], presenting the A4L2.0 architecture, identified the problem directly: "learner data is distributed across multiple, and often disparate, systems," and proposed a data pipeline built on open standards. A4L2.0 demonstrates that the integration gap is real and technically addressable, but it is a bespoke infrastructure solution rather than a diagnostic instrument for evaluating what a platform already provides. The IEEE xAPI standard [22] defines a structured format for recording learning experiences across systems, but xAPI describes a data exchange protocol, not a platform readiness assessment. Kaleci [23], examining AI integration in Moodle, documented configuration-level integration using plugins and APIs but evaluated readiness at the level of installed features rather than the underlying event and data architecture that agents would need to operate autonomously.

2.3 The Gap STAIR Addresses

Prior readiness frameworks assess adjacent dimensions. Capability Maturity Model Integration (CMMI) and Technology Readiness Level (TRL) measure organisational process and technology maturation. The Substitution-Augmentation-Modification-Redefinition (SAMR) model and the Technological Pedagogical Content Knowledge (TPACK) framework measure pedagogical integration and teacher knowledge. General AI-readiness matrices measure organisational capability for AI adoption. None of them asks whether a platform is architecturally capable of hosting an autonomous teaching agent. Without diagnosing architectural readiness, educational AI projects can fail silently at the platform boundary while model quality and pedagogical design receive all the scrutiny. STAIR is built to surface that failure before it happens. Section 3 derives its five levels from self-regulated learning theory, multi-agent collaboration research, and the educational large language model (LLM) agent literature. The resulting rubric scores whether a learning management system can structurally host an autonomous teaching agent.

3. The STAIR Framework

STAIR (Scaffold for Teaching Agent Integration Readiness) is a five-level diagnostic framework. It evaluates whether a learning management system provides the infrastructure an autonomous AI teaching agent needs to sense, reason, coordinate, and act. Each level answers one question. Can an agent function without this capability? If the answer is no, and no existing level already captures the requirement, it defines a new level. The five levels depend on each other in strict sequence. Without event sensing (L1), student-state

modelling (L2) has no input; without reasoning (L3), orchestration (L4) has nothing to coordinate. This structure converges across the three theoretical pillars. Zimmerman's SRL cycle, Talebirad and Nadiri's multi-agent collaboration framework, and Chu et al.'s educational LLM agent taxonomy each independently name requirements that map onto the same five levels, evidence that each boundary reflects a qualitative transition rather than an arbitrary division. Figure 2 presents the full architecture and Table 1 defines each level.

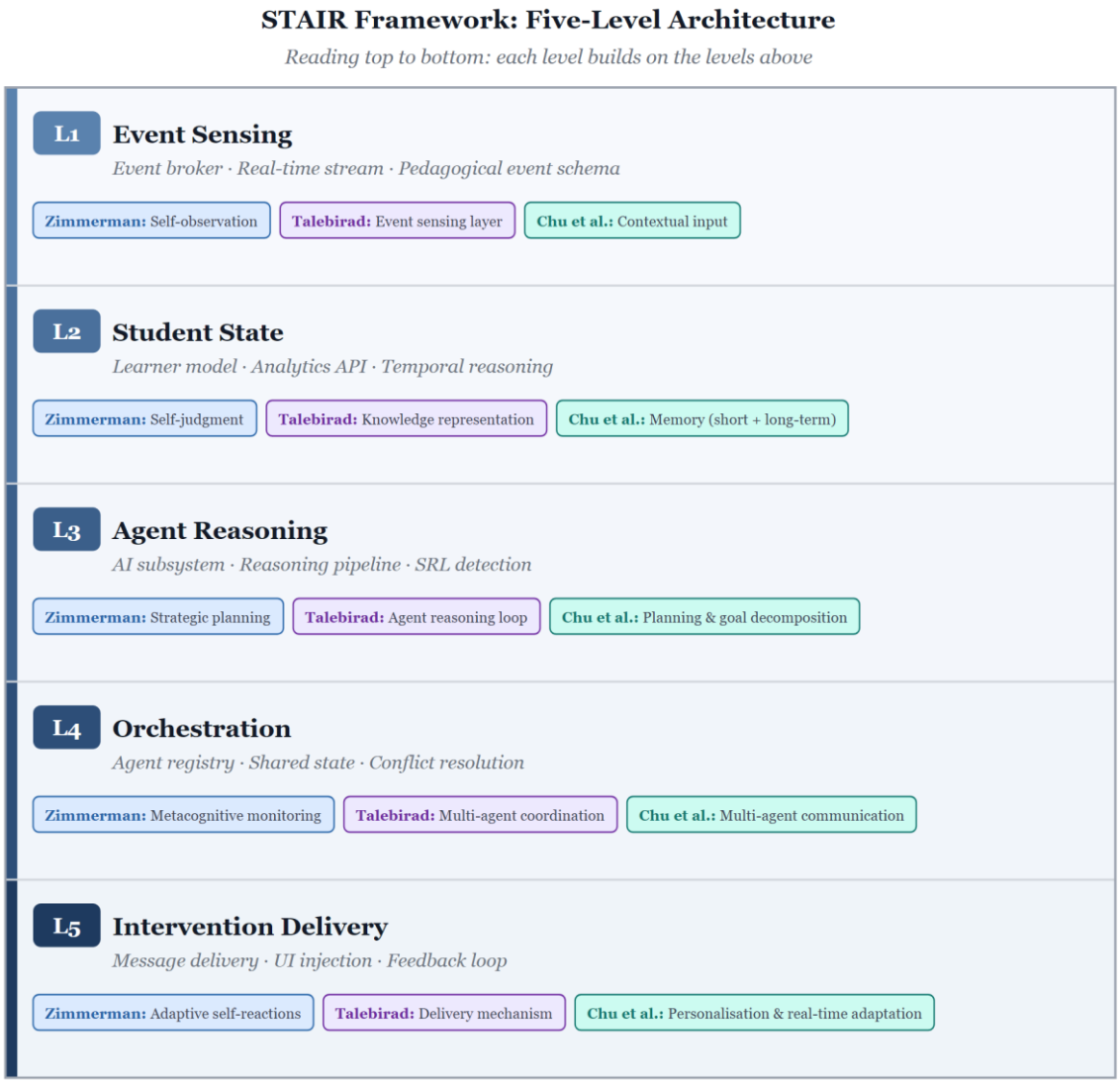

FIGURE 2. STAIR five-level architecture with dependency chain and theoretical grounding from Zimmerman (SRL), Talebirad and Nadiri (multi-agent collaboration), and Chu et al. (educational LLM agents).

TABLE 1. STAIR framework levels with architectural requirements and illustrative examples.

STAIR Level	Architectural Requirement	Illustrative Example
L1: Event Sensing	Platform emits structured, pedagogically classified learner events accessible to AI consumers.	The platform records that a student has abandoned three consecutive quiz attempts within one hour.
L2: Student State	Platform maintains a persistent, queryable model of each learner's cumulative progress, performance, and risk status.	The agent queries the platform and finds the student's completion rate has fallen from 85% to 40% over two weeks.
L3: Agent Reasoning	Platform supports autonomous AI deliberation over student state, with platform-enforced pedagogical guardrails.	The agent analyses the declining completion, the risk flag, and the abandonment pattern, then selects a scaffolded hint strategy.
L4: Orchestration	Platform coordinates concurrent AI agents sharing student context, with conflict resolution and load management.	A tutoring agent and a motivation agent both receive the student's state; the orchestration layer sequences their outputs.
L5: Intervention Delivery	Platform delivers context-aware, timely agent interventions embedded within the flow of learning activity.	As the student opens the quiz for a fourth attempt, the platform injects a motivational prompt and a targeted hint.

Note. Levels represent necessary conditions in a strict dependency chain. A platform that fails at L1 cannot support L2 regardless of its analytics capabilities, because agents have no perceptual input.

3.1 Level 1: Event Sensing

An agent that cannot observe the learning environment has no input from which any further function can proceed. The platform must emit structured, classifiable events carrying sufficient metadata (actor, context, action, timestamp) for an agent to distinguish pedagogically relevant actions from administrative ones. Zimmerman [1] identifies self-observation as what feeds self-regulation, making observable behaviour the dependency both the learner and an external agent rely on.

3.2 Level 2: Student State

Events describe individual moments; effective teaching requires understanding trajectories. The platform must maintain a persistent, queryable model of each learner's grades, completion, engagement patterns, and risk indicators. This level is categorically distinct from event sensing: L1 concerns the flow of individual observations; L2 concerns their aggregation into a coherent learner model. Zimmerman [1] draws the same boundary, distinguishing self-observation from self-judgment as different cognitive operations requiring different architectural support.

3.3 Level 3: Agent Reasoning

A platform with complete learner data but no reasoning infrastructure cannot translate knowledge into a pedagogically appropriate response. The platform must support autonomous agent deliberation over learner state, including LLM access, learner context input, and platform-enforced educational constraints bounding the agent's action space. Talebirad and Nadiri [18] show that agents maintain an evolving reasoning process operating over stored state, establishing reasoning as an active deliberation layer rather than a retrieval operation.

3.4 Level 4: Orchestration

Specialised agents addressing cognitive, motivational, and strategic demands must be coordinated. Contradictory interventions undermine each other and may harm learning outcomes. The platform must provide agent registration, shared learner state accessible to concurrent agents, and arbitration mechanisms preventing conflicting interventions. Talebirad and Nadiri [18] demonstrate that multi-agent systems require shared workspaces and supervisory agents, establishing coordination as qualitatively distinct from single-agent reasoning.

3.5 Level 5: Intervention Delivery

Correct reasoning and coordinated agents do not guarantee educational impact if the intervention cannot reach the learner at the moment it is needed. The platform must enable agents to deliver context-aware support embedded within the learning activity, through appropriate channels, at pedagogically appropriate moments. Zimmerman [1] identifies adaptive self-reactions as the output that catalyses the next SRL cycle, emphasising that timing and context determine whether support activates the learner's own regulatory processes.

4. Methodology

4.1 Research Design

This study adopts a design science paradigm [24], following the design science research methodology [25]. Two artifacts are produced: a diagnostic framework (STAIR, a model artifact) and a scoring instrument (a method artifact) that operationalises the framework into reproducible assessment procedures. The problem motivating this design is the absence of any instrument for evaluating whether a learning management system can architecturally support autonomous AI teaching agents.

4.2 Case Selection

Moodle 5.2dev+ (Build: 20260320) was selected for three reasons: it is the most widely deployed open-source LMS globally, its 5.x branch introduced a dedicated AI subsystem (core_ai) making it the first release with native AI integration, and full source code availability enables direct verification of every architectural claim against the implementation.

4.3 Data Collection

Data were collected through structured source code analysis of the complete Moodle 5.2dev+ repository. The analysis followed a four-tier evidence hierarchy in decreasing evidential strength: source code inspection (S-type), API documentation and web service definitions (A-type), runtime observation (R-type), and official developer documentation (D-type). S-type evidence, the primary method, was obtained for all 28 sub-dimensions through systematic inspection of subsystem directories. A-type evidence supplemented S-type findings for externally documented capabilities. R-type and D-type evidence were used for confirmation but were not required as primary justification for any score. All evidence strings are recorded in `analysis/data/scoring_provenance.json` in the public reproduction repository (see Code and Data Availability), mapping each score to the Moodle source files, classes, and constants that justify it. An interactive browser of the complete rubric, sub-dimension definitions, evidence, Table 3 derivation, architecture finding, and event catalog is available at `docs/methodology_explorer.html` in the same

repository, giving a non-technical reader a single self-contained walkthrough from raw evidence to final score.

4.4 Scoring Instrument

The STAIR Scoring Instrument v1.0 operationalises the five framework levels into 28 sub-dimensions distributed across those levels (L1: 5, L2: 6, L3: 7, L4: 5, L5: 5). Each sub-dimension is scored on a four-point ordinal scale: 0 (Absent, 0%), 1 (Nascent, 25%), 2 (Partial, 50%), 3 (Substantial, 75%), 4 (Ready, 100%). Rubric boundaries are defined by explicit evidence criteria: a score of 2 requires at least one S-type and one A-type evidence item for externally facing capabilities; sub-dimensions describing purely internal architecture (e.g., cross-subsystem integration) are scored on S-type evidence alone. A score of 0 requires an S-type search confirming absence.

Sub-dimension scores are mapped to five consolidated capability dimensions: Coverage (COV), External Accessibility (ACC), Educational Specificity (EDU), Temporal Availability (TMP), and Cross-Subsystem Integration (INT). These five dimensional scores are averaged to produce the level readiness percentage. Equal weighting was chosen because no empirical basis currently exists for differential weighting; assigning higher weight to one dimension would require evidence from agent deployment outcomes that the field has not yet produced. Sensitivity analysis varying these weights is identified as future work once multi-platform data become available. Figure 3 illustrates the complete scoring methodology flow from theoretical grounding through empirical application to the final readiness profile. The complete sub-dimension breakdown, scoring computations, and source-code evidence provenance are available in the supplementary materials.

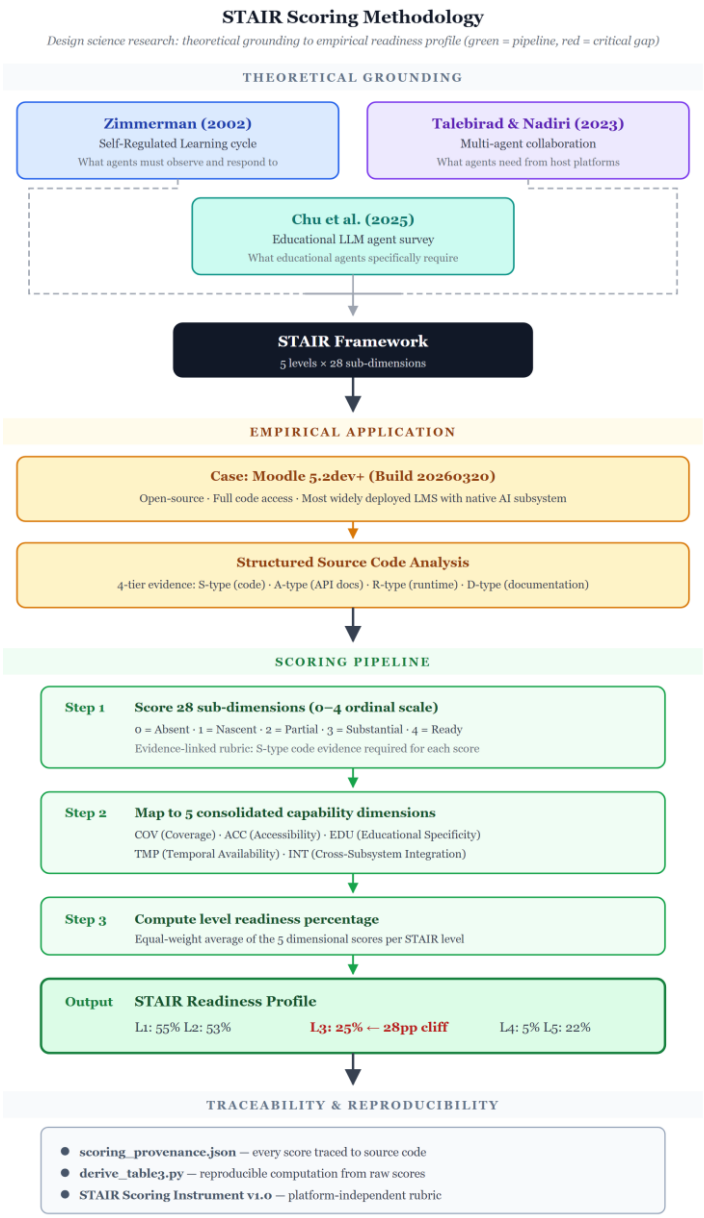

FIGURE 3. STAIR scoring methodology flow: theoretical grounding through empirical application to readiness profile. Three-step scoring pipeline maps 28 sub-dimensions to five consolidated capability dimensions and averages them into per-level readiness percentages.

4.5 Limitations

Four limitations bound the scope of these findings. The instrument's dimensional structure is derived from theoretical convergence rather than empirical factor analysis; alternative weighting schemes could produce different level scores, and empirical calibration is identified as future work once multi-platform data become available. The evaluation was conducted by a single rater without inter-rater reliability assessment; the instrument was designed for reproducibility, but independent replication is warranted. The analysis covers a single platform at a single point in time; scores reflect Moodle 5.2dev+ and will change as the platform evolves. The evaluation analysed source code rather than deployed system behaviour; runtime performance characteristics such as event latency under load were not measured.

5. Results

Moodle 5.2dev+ exhibits a structural cliff. The readiness profile falls 28 percentage points between the data levels and the reasoning level, not gradually but as a discontinuity. Figure 4 presents the full score pattern across all 28 sub-dimensions. The platform scores 55% at L1 (Event Sensing) and 53% at L2 (Student State), falls to 25% at L3 (Agent Reasoning), and remains low at L4 (5%) and L5 (22%). The L1–L2 plateau reflects mature data collection and access. The drop at L3 and the patterns at L4 and L5 reflect a separate architectural condition — the absence of reasoning, coordination, and timing infrastructure that autonomous agents require. Table 3 decomposes each level score into five consolidated dimensions (coverage, external accessibility, educational specificity, temporal availability, and cross-subsystem integration, abbreviated COV, ACC, EDU, TMP, and INT).

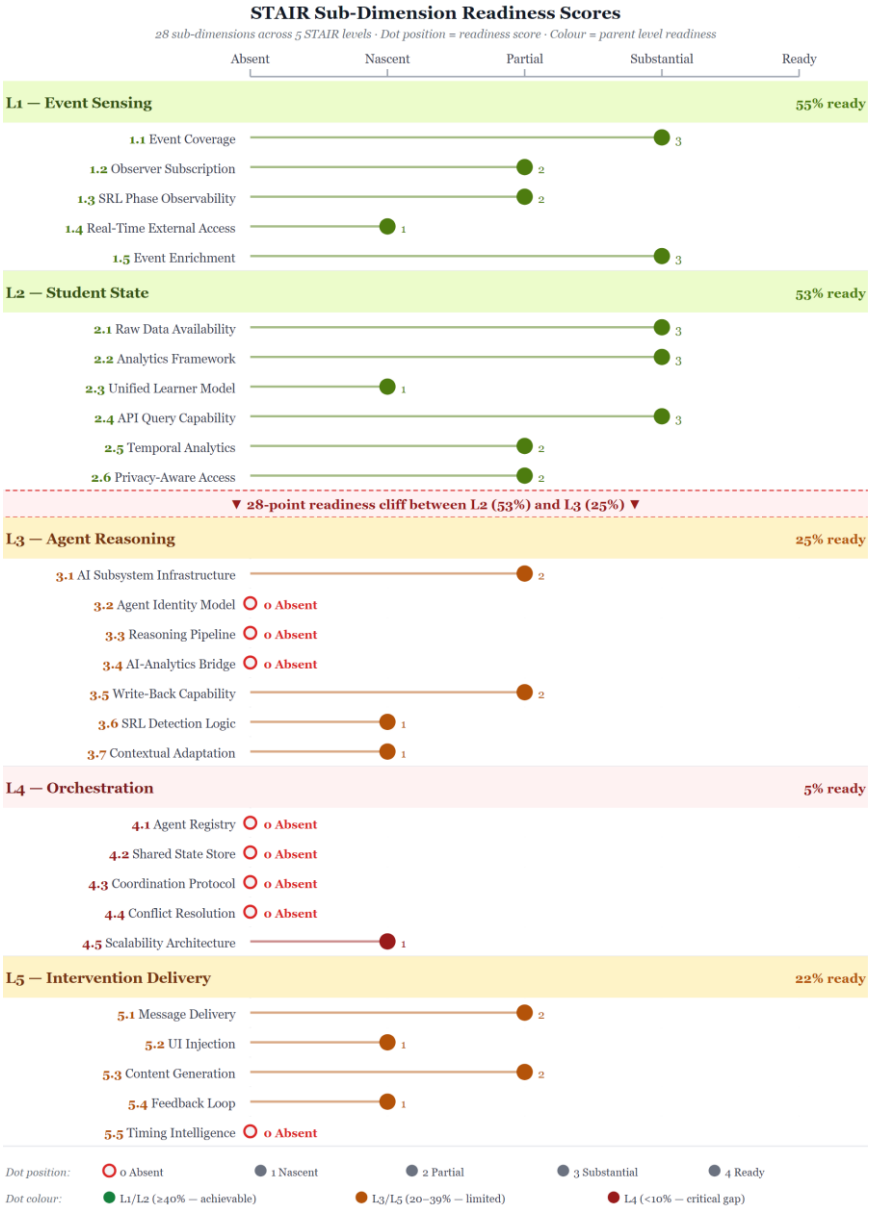

FIGURE 4. STAIR sub-dimension readiness scores across all 28 dimensions. The red band highlights the 28-percentage-point discontinuity between L2 (53%) and L3 (25%); colour encodes the parent level's readiness band.

TABLE 3. Sub-dimension scoring matrix for Moodle 5.2dev+ across five consolidated capability dimensions.

STAIR Level	COV	ACC	EDU	TMP	INT	Level Avg.
L1: Event Sensing	75	50	50	25	75	55%
L2: Student State	75	75	25	50	38	53%
L3: Agent Reasoning	50	50	0	25	0	25%
L4: Orchestration	0	0	0	0	25	5%
L5: Intervention Delivery	50	25	12	0	25	22%

Note. Cells consolidate 28 sub-dimensions into five capability dimensions (COV = Coverage and Classification; ACC = External Accessibility; EDU = Educational Specificity; TMP = Temporal Availability; INT = Cross-Subsystem Integration). Each cell is the weighted mean of its contributing sub-dimensions; where two jointly define a dimension, each carries half-weight. L3's seven sub-dimensions populate its row as: COV = 3.1 (1.0); ACC = 3.5 (1.0); EDU = 0.5 × 3.2 + 0.5 × 3.3; TMP = 0.5 × 3.6 + 0.5 × 3.7; INT = 3.4 (1.0). Raw 0-4 scores convert to 0/25/50/75/100 before averaging. Per-cell arithmetic in `docs/methodology_explorer.html`; weights in `analysis/data/scoring_provenance.json`. Moodle 5.2dev+ (Build 20260320).

5.1 L1: Event Sensing (55%)

L1 scored 55% because the platform classifies learning actions well but cannot stream them in real time. The 228 core event classes each carry an educational-level taxonomy (edulevel: student-participating, teaching, or administrative), so a consuming agent can distinguish learning actions from administrative ones without parsing raw event content. This classification produces COV = 75%. Mapping the event stream onto Zimmerman's three SRL phases reveals an uneven distribution. Performance-phase events are well covered, but forethought-phase events (goal-setting, planning) are uniformly weak. An agent designed to scaffold goal-setting therefore has almost no perceptual signal at this layer.

Real-time external access is the L1 bottleneck (TMP = 25%). Moodle dispatches events only to code running inside its own PHP codebase. No WebSocket, webhook, or message queue exposes them to external listeners, so an outside agent must poll the database log for changes. The resulting delay makes in-session response to a student's unproductive struggle impossible.

5.2 L2: Student State (53%)

L2 scored 53% because the platform exposes extensive learner data but fragments it across four subsystems and locks risk predictions behind an internal-only interface. Grades, completion progress, competency evidence, and dropout-risk predictions are all stored in queryable tables (COV = 75%), and 402 core web service functions make courses, completion, grades, enrolment, and competencies accessible to authenticated external clients (ACC = 75%), representing one of the broadest programmatic data interfaces among major learning platforms.

Two dimensions limit the usability of this breadth for agent consumers. The educational specificity dimension (EDU = 25%) captures a fragmentation problem. Student state is distributed across separate grade, completion, competency, and analytics subsystems with no unified query point, so an agent assembling a complete learner profile must issue at least four independent API calls and accept no guarantee of temporal consistency. The integration dimension (INT = 38%) captures a predictions gap. Analytics predictions are stored in an internal database table with no corresponding web service endpoint, so an agent cannot programmatically query whether a particular student is at risk. The temporal dimension (TMP = 50%) compounds both limits. Predictions are generated through a scheduled batch process with no on-demand trigger, and a deteriorating engagement pattern may not be reflected in a queryable score until a substantial interval has elapsed.

5.3 L3: Agent Reasoning (25%)

The L2-to-L3 transition is the paper's central finding. The 28-percentage-point drop is produced not by a single weak sub-dimension but by a structural condition that forces zero scores on two of the five consolidated dimensions.

The integration dimension scored 0% because the AI subsystem and the analytics framework share no code pathway. The AI subsystem (ai/) contains no imports from the analytics namespace, the analytics framework (analytics/) contains no imports from the AI namespace, and neither subsystem references the other's database tables or event channels. This is not a configuration gap. The two subsystems were developed independently and share no data pathway at all. The platform can generate AI responses, and the platform can identify struggling students, but these capabilities operate as entirely separate systems. Figure 5 illustrates the isolation and shows what a connecting layer would require.

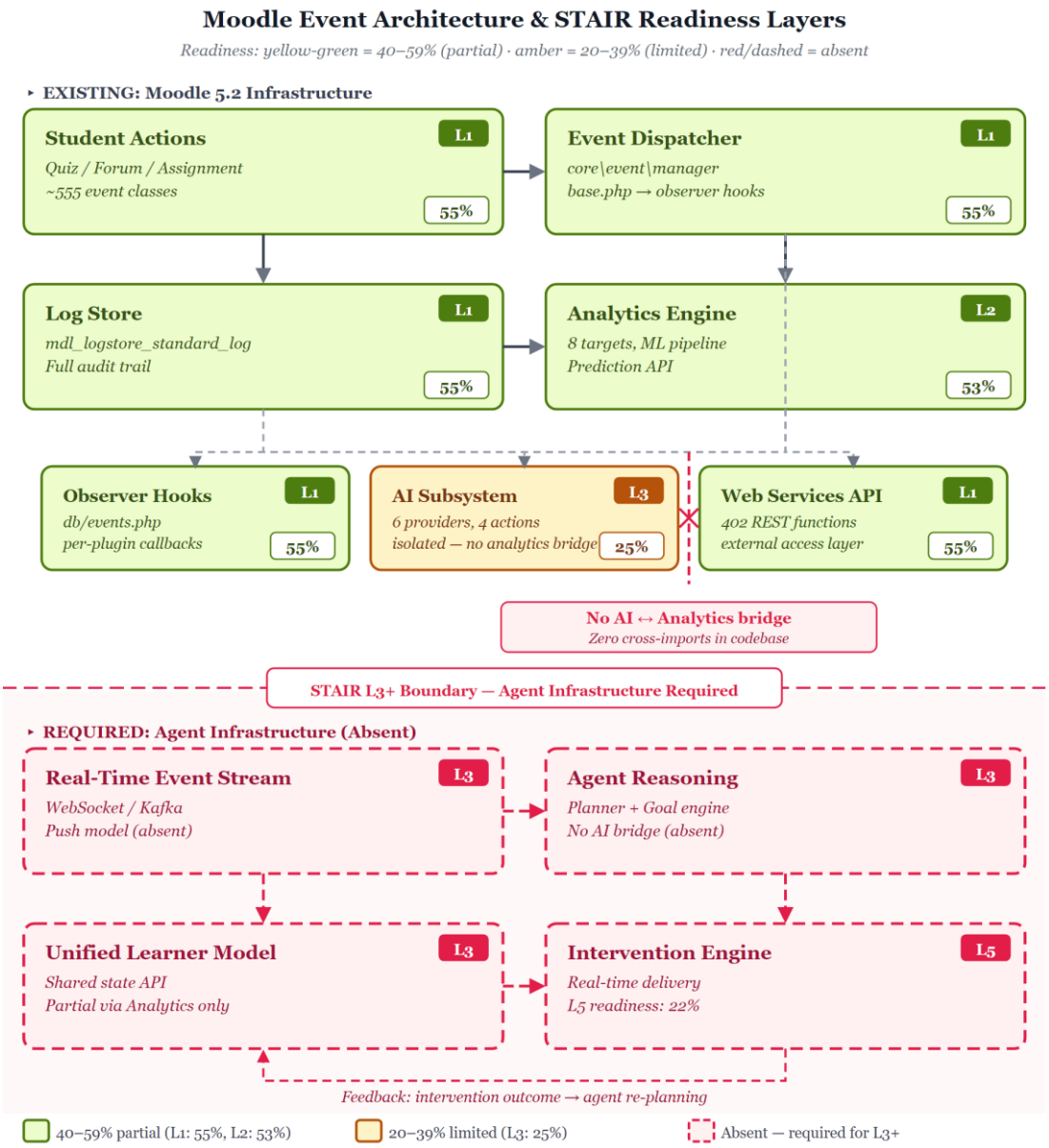

FIGURE 5. Moodle event architecture mapped to STAIR readiness levels. Solid boxes represent existing infrastructure; dashed red boxes represent absent infrastructure required for L3+. The central band highlights the AI–analytics isolation producing the L2-to-L3 cliff.

The educational specificity dimension also scored 0%, reflecting the complete absence of agent reasoning infrastructure. No agent role exists in the platform's capability system, so agents cannot be assigned distinct permissions or identified as a class of actor separate from human users. The platform contains no reasoning pipeline, no goal decomposition mechanism, and no facility for an AI action to initiate a subsequent action in response to a student state change. Each AI action processes as an independent, stateless request-response pair with no memory of prior interactions.

The coverage dimension (COV = 50%) confirms that the gap is architectural, not technical. Six LLM provider integrations with rate limiting, audit logging, and configurable model selection constitute a strong technical foundation for language model access. The missing piece is the bridge that would connect that capability to learner context. An AI subsystem with access to six language models but no access to student performance data cannot produce a pedagogically grounded response.

5.4 L4 Orchestration (5%) and L5 Intervention Delivery (22%)

L4 at 5% reflects a frontier condition shared across learning platforms, not a Moodle-specific deficiency. Four of the five dimensions scored 0% (no agent registry, no shared agent state store, no coordination protocol, no real-time coordination mechanism). The integration dimension scored 25% because the scheduled task infrastructure provides repurposable scheduling primitives without agent-aware coordination logic. Multi-agent orchestration requires three preconditions that no major learning platform currently provides: an agent identity model, a shared learner state store, and a functioning agent reasoning layer.

L5 at 22% reflects a distinct pattern. Delivery infrastructure exists in functional form, but the intelligence required to make it pedagogically useful is absent. The coverage dimension (COV = 50%) captures the functional side. The messaging API can send messages to individual users or conversation groups, and the

AI content generation pipeline supports text generation, summarisation, and explanation through six LLM providers. The temporal and educational specificity dimensions (TMP = 0%, EDU = 12%) capture the absent intelligence. No timing engine detects that a student is in a moment of difficulty, and no intervention tracking links a delivered message to subsequent student behaviour. A platform that can send a message but cannot determine when to send it, to whom among those currently struggling, or whether previous messages helped, does not satisfy the conditions for agent-mediated intervention delivery. L5 exceeds L4 because delivery infrastructure was developed independently for teacher use and is programmatically accessible to agents. Section 6 examines why this data-reasoning cliff is the structural condition these results identify, and what a bridge across it would require.

6. Discussion

6.1 Central Finding and Mechanism

The L2-to-L3 discontinuity is an architectural boundary, not a performance gap. Moodle's event and student state infrastructure was built for human consumers. Teachers review gradebooks, administrators generate reports, and researchers export datasets. These are human workflows the platform handles well. Autonomous agents need something different. They need real-time perception rather than retrospective queries, integrated learner models rather than isolated records, and persistent reasoning over accumulated context rather than stateless request-response. The 28-percentage-point drop at L3 reflects this mismatch between human-oriented infrastructure and agent-oriented demands, not a deficiency in Moodle-specific engineering.

The mechanism is code-level isolation between the AI subsystem and the analytics framework. The two subsystems share no imports, no database tables, and no event subscriptions. The platform can generate a language model response, and the platform can predict student risk, but no architectural pathway connects these capabilities. Any agent operating within the current platform must construct its own bridge between language model access and learner context.

6.2 Bridge Feasibility and Practical Implications

The L3 isolation is addressable without modifying Moodle's core. Analytics predictions are exposed via `get_predictions` and AI action subclasses receive configuration via `get_configuration`, so a local plugin bridging these two interfaces could inject predictions into the AI action context at roughly 150–200 lines of code, an estimate drawn from source analysis that remains to be validated empirically. For researchers, STAIR replaces the binary "supports AI" claim with five architectural levels, distinguishing AI-assisted analysis from autonomous teaching agents.

6.3 Generalisability and Broader Significance

STAIR's 28 sub-dimensions derive from the convergence of three theoretical frameworks [1], [18], [19], not from any single platform. Its evidence types and boundary criteria therefore transfer to other LMS platforms without modification. The multi-platform evaluation identified in Section 7 as the most important next step will test this prediction on Open edX and Canvas first. The field has long recognised that educational AI depends on platform infrastructure, but has lacked a structured way to measure that dependence. STAIR converts this problem into a common five-level scale on which platforms can be compared, architectural weaknesses prioritised, and "AI-ready" claims evaluated against specific, evidence-anchored levels.

6.4 Limitations

Three limitations bound these conclusions. The evaluation was conducted by a single rater without inter-rater reliability assessment. The instrument was designed for reproducibility, but independent replication is warranted. The analysis covers a single platform at a single point in time, so Moodle's architectural patterns may not generalise to platforms built on different technology stacks. The evaluation analysed source code rather than runtime behaviour, and a capability present in the codebase may not function as intended in deployment. STAIR assesses structural readiness rather than operational readiness.

7. Conclusion

This paper contributes three instruments to a field that has lacked them. First, the STAIR framework operationalises what "agent readiness" means at the platform level. It is a five-level diagnostic grounded in the converging requirements of self-regulated learning theory, multi-agent collaboration architecture, and educational LLM agent research. Second, the first source-code-based application of STAIR to Moodle produced an empirical readiness profile revealing a 28-percentage-point structural collapse between the data levels and the reasoning and coordination levels. Third, the evaluation identified the mechanism behind this discontinuity. Complete code-level isolation between the AI subsystem and the analytics framework creates a barrier that is architectural within the platform core but bridgeable through the plugin system without core modification.

The most important next step is multi-platform evaluation applying STAIR to other open-source learning management systems. Open edX (Python/Django with its own AI integration layer) and Canvas LMS (REST-first architecture) are prioritised comparison cases. A comparative study would test whether Moodle's

readiness patterns generalise across platforms built on different technology stacks and produce the inter-rater reliability data needed to validate the scoring instrument empirically. Secondary priorities include longitudinal re-evaluation of Moodle as the AI subsystem matures, and sensitivity analysis varying the currently uniform dimension weights once multi-platform data become available.

The broader question is whether the field's investment in educational AI research can be realised at institutional scale without corresponding investment in platform architecture. The capability gap identified here is not in the AI; it is in the infrastructure that must host it. Until learning management systems are designed to support perception, reasoning, coordination, and intervention at the architectural level, the promise of autonomous AI teaching agents will remain constrained by the platforms hosting them.

Declaration on Generative AI

During the preparation of this work, the author used Claude (Anthropic) for the following purposes: grammar and spelling checks, paraphrasing and rewording, and generating analysis and visualisation code. All empirical claims about Moodle's architecture were independently verified against the Moodle 5.2dev+ (Build: 20260320) source code, and all scoring decisions, interpretations, and conclusions are the author's own. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the publication's content.

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Code and Data Availability

The analysis scripts, scoring instrument, rubric data, and reproduction harness for this paper are available at https://github.com/baqarjafri/paper-stair-framework under MIT (code) and CC-BY-4.0 (paper and figures). The Moodle source code analysed is Moodle 5.2dev+ at commit 213a869e7f, browsable at https://github.com/moodle/moodle/tree/213a869e7ffd48d3a5679818bcddbb22dbc31aa5. Running *python reproduce.py* fetches this exact commit and regenerates every quantitative claim and figure in this paper.

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